A SYSTEM AND METHOD FOR HIGH-RADIX PARAMETRIC OPTICAL GROUND-TO-SPACE COMMUNICATION

Reinaldo Javier Aguiar Marcano

April 7, 2026

Title: A SYSTEM AND METHOD FOR HIGH-RADIX PARAMETRIC OPTICAL GROUND-TO-SPACE COMMUNICATION

Inventor: Reinaldo Javier Aguiar Marcano, a citizen of the United States, residing at 2302 Britton Ridge Drive, Katy, TX 77494.

Cross-Reference to Related Applications

This application builds upon and extends the principles described in prior applications concerning High-Radix Number Representation, Parametric Networking, and High-Bandwidth Optical Data Cables, the contents of which are incorporated by reference.

Abstract

A system and method for secure, high-bandwidth, one-way data communication from a ground station to an orbiting satellite. The system comprises a ground-based array of physically-shielded, individually-addressable, multi-wavelength light emitters. A control system modulates the emitters to create a large-scale optical pattern, wherein the data is encoded using high-radix and parametric encoding schemes, mapping complex data "recipes" to the state of each emitter. The resulting optical signal is transmitted vertically to a satellite equipped with an optical sensor payload. The physical shielding of each emitter collimates

the light and makes the array difficult to detect from ground level or non-overhead angles, providing a high-security, high-bandwidth optical uplink that is resistant to interception.

Background of the Invention

Satellite communication uplinks have traditionally relied on radio frequency (RF) technology. RF uplinks are subject to several limitations, including the need for spectrum licensing, significant power requirements, and vulnerability to both interception and jamming by third parties. While free-space optical (FSO) communication has been proposed as an alternative, existing FSO systems still predominantly rely on binary encoding schemes and do not fully exploit the information-carrying capacity of light. Furthermore, a high-power FSO transmitter can be easily detected. There is a need for a ground-to-space communication system that offers massive bandwidth and a low probability of interception.

Summary of the Invention

The present invention is a system for optical ground-to-space communication that achieves unprecedented bandwidth and security by combining a novel physical array design with advanced, high-cardinality data encoding methods.

The system consists of a ground station and a satellite payload. The ground station features a large, planar array of light-emitting units. Each unit is individually addressable by a central controller and is physically shielded by deep, opaque baffles that collimate the light into a narrow, vertical beam. This shielding prevents light leakage, making the array extremely difficult to observe from any ground-level or off-axis position. The controller encodes data using high-radix, parametric schemes, where the specific wavelength and intensity of a single emitter can represent a complex mathematical recipe for a number, rather than a simple binary digit.

An orbiting satellite equipped with an optical sensor payload is timed to pass over the ground station. During the overpass, the ground array transmits a sequence of frames, creating a massively parallel optical data stream. The satellite's sensor captures these frames. Because the communication is a highly-collimated, one-way uplink, interception of the signal would require an adversary to be positioned in space, directly between the ground station and the satellite, at the exact time of transmission, which is practically infeasible. This results in a secure, ultra-high-bandwidth data uplink.

Brief Description of the Drawings

FIG. 1 is a system diagram illustrating the ground array transmitting the encoded optical uplink to the orbiting satellite.

FIG. 2 is a detailed cross-sectional view of a single emitter unit, illustrating the light source and the opaque shielding.

Detailed Description of the Invention

Referring to **FIG. 1**, the overall system for optical ground-to-space communication is shown. A ground station (100) comprises a large, planar array (101) composed of thousands of individual emitter units. A controller module (not shown) sends data to the array, which generates a high-radix, parametrically encoded optical uplink (104). The uplink consists of a structured pattern of light with varying wavelengths and intensities. The signal is directed vertically towards a designated orbital path. An orbiting satellite (105), equipped with a high-sensitivity optical sensor (106), passes over the ground station and captures the sequence of optical frames transmitted by the array, thereby receiving the data.

Referring to **FIG. 2**, a detailed view of a single Ground Emitter Unit (200) from the array is shown. The unit is designed for security and signal integrity. At its base is a multi-wavelength light source (201), such as a high-intensity RGB LED or a tunable laser diode. The light source is positioned at the bottom of a deep, tube-like structure made of an opaque material, which forms the baffles or shielding (203). This structure acts as a collimator, forcing the light to travel in a narrow, vertical beam. This physical shielding ensures that the light emission is not visible from ground level and is only detectable by an overhead observer, drastically reducing the probability of interception. An optional collimating lens (202) may be placed above the light source to further focus the beam.

The data encoding itself leverages the principles of high-radix and parametric representation. A single emitter's state (defined by its wavelength, intensity, and potentially polarization) is not mapped to a binary 1 or 0, but to a high-cardinality symbol that can represent a complex "recipe" for data. With an array of over 100,000 such emitters on a one-acre plot, refreshing at a high frame rate, the aggregate data uplink bandwidth can be enormous, reaching terabits per second.

Claims

What is claimed is:

1. A system for ground-to-space data communication, comprising:

    - a ground station comprising a planar array of a plurality of individually-addressable light emitter units;
    - wherein each light emitter unit comprises a light source housed within an opaque, tubular shielding structure that collimates emitted light into a substantially vertical beam and reduces observability from off-axis ground-level positions; and
    - a control module configured to modulate the light sources in the array to produce an optical signal encoded with data for transmission to an orbiting satellite.

2. The system of claim 1, further comprising an orbiting satellite equipped with an optical sensor payload configured to receive the optical signal.

3. The system of claim 1, wherein the control module is configured to encode the data using a high-radix, parametric encoding method, wherein the state of a single light emitter unit represents a multi-parameter data structure.

4. The system of claim 1, wherein each light emitter unit is a multi-wavelength light-emitting diode (LED).

5. The system of claim 1, wherein the shielding structure is a cylindrical tube or baffle.

6. A method for transmitting data from the ground to a satellite, comprising the steps of:

    - providing a ground-based array of individually-addressable light emitters, wherein each emitter is housed within a shielding structure to collimate its light output;
    - encoding a data stream into a sequence of optical frames, wherein the data is represented by the state of the light emitters in the array;
    - transmitting said sequence of optical frames from the array as a directed optical signal towards a predicted position of an orbiting satellite; and
    - receiving said optical signal at an optical sensor on the orbiting satellite.

7. The method of claim 6, wherein the step of encoding comprises using a high-radix, parametric scheme to map the data to the wavelength and intensity of the light emitters.

8. The method of claim 6, wherein the shielding structure reduces the observability of the array from ground level.

Drawings

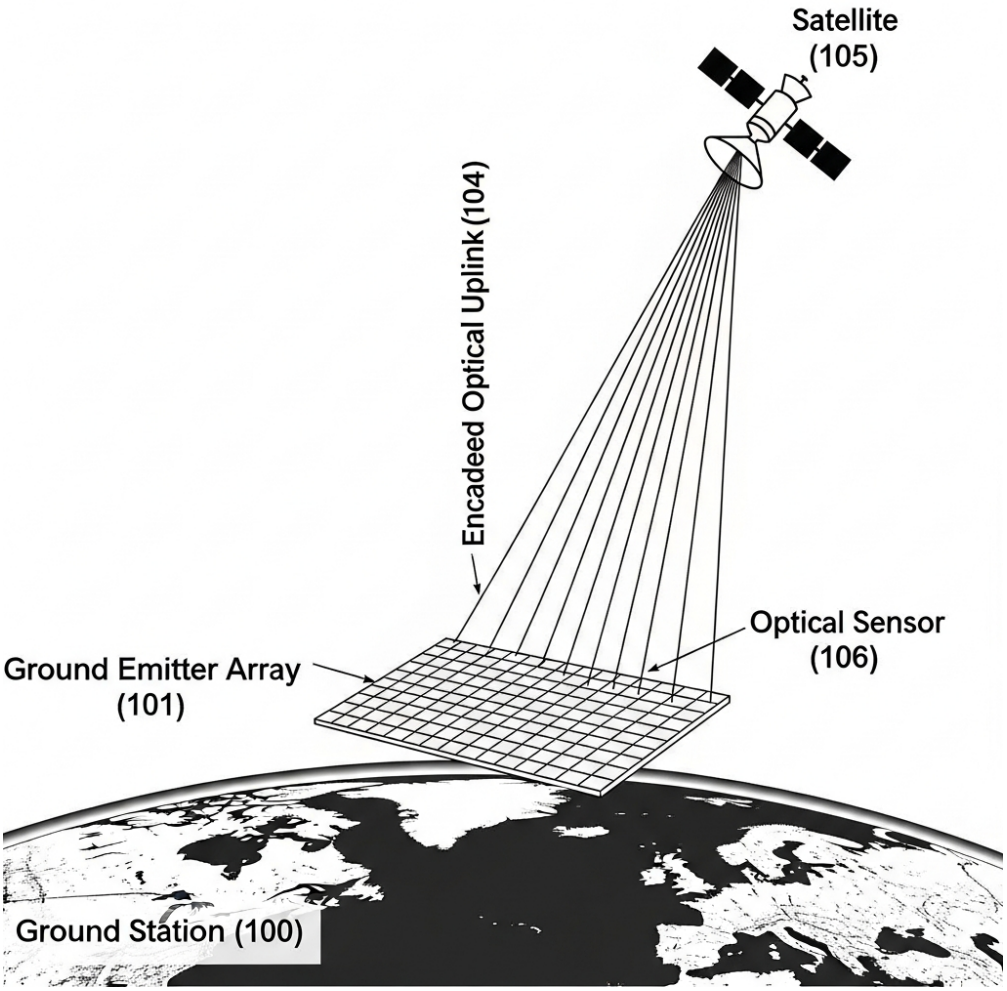


Figure 1: System diagram illustrating the ground array transmitting the encoded optical uplink to the orbiting satellite.

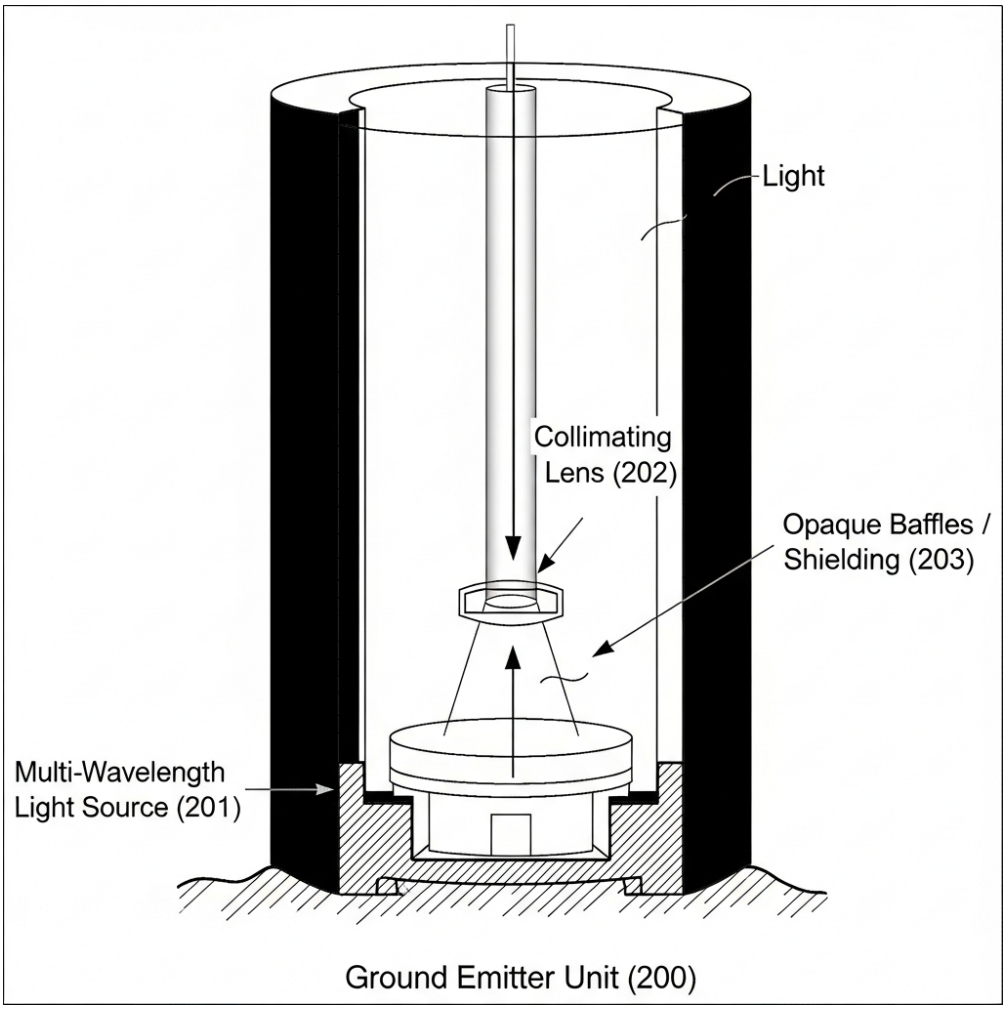


Figure 2: Detailed cross-sectional view of a single emitter unit, illustrating the light source and the opaque shielding.